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Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738066522388-Chapter 4_unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

TWINKLE TUNES: EUN'S MUSICAL DISCOVERY

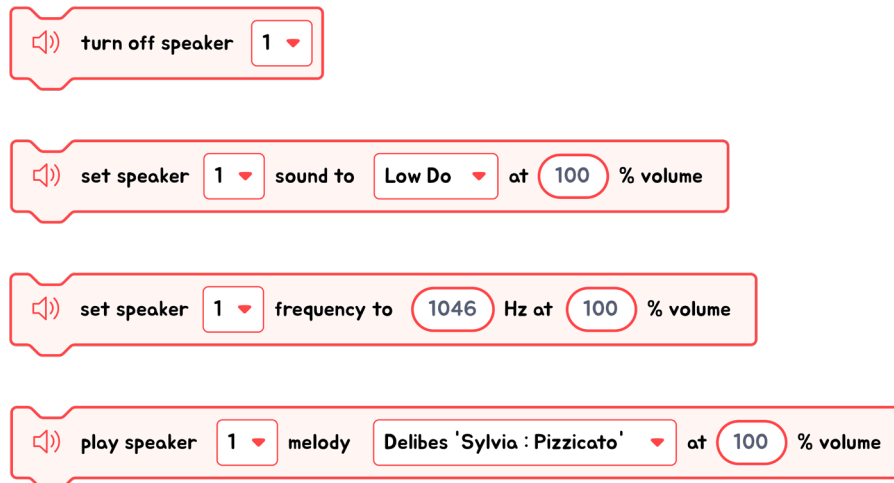
Meet Eun, a creative seventh grader who loves all things music and tech. One sunny afternoon, while digging through her family's old storage boxes in the attic, Eun found an intriguing box filled with various electronic parts. Among these were colorful modules, wires, and some buttons. The sight of these components sparked an idea in Eun's mind: why not build something that combines her love for music with her interest in technology? Excited by the challenge, Eun imagined creating a music box that could play her favorite lullaby, "Twinkle Twinkle Little Star."



Eager to share her idea and get some help, Eun called her friends, Cathy and Philipp. They loved the idea and agreed to help bring the music box to life. Together, the trio spent several afternoons after school working on the project. Eun and her friends designed a music box equipped with a button that, when pressed, plays "Twinkle Twinkle Little Star." The music box combines a storage mechanism for the tune and an output system for sound delivery, all integrated into a user-friendly and aesthetically pleasing device. This setup ensures each button press reliably activates the beautifully programmed melody, showcasing an effective blend of electronic hardware with creative software solutions.



ABOUT SPEAKER MODULE

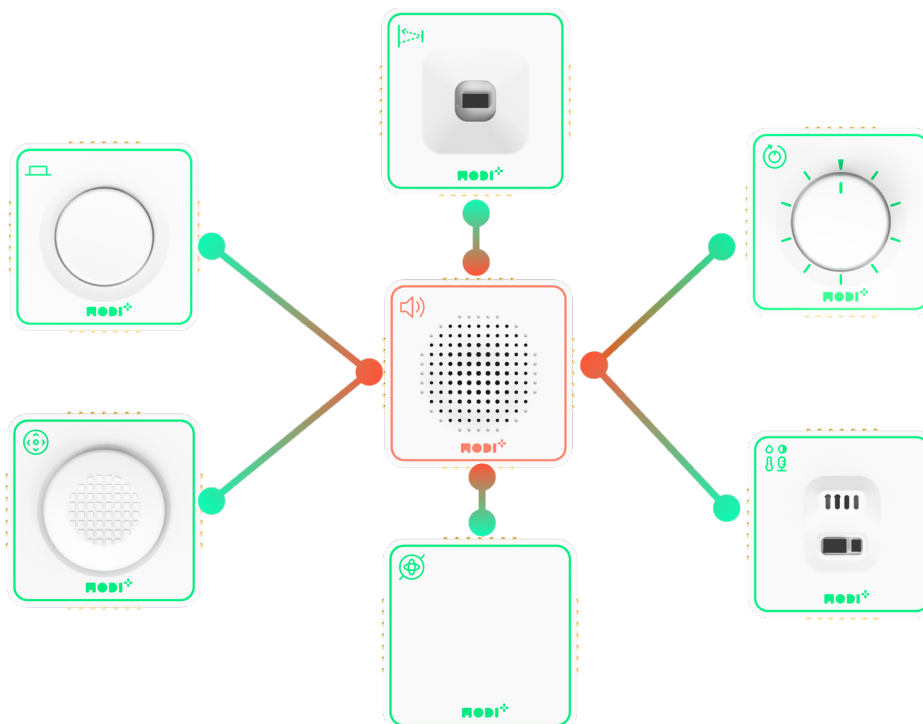


In the Code Editor, the Speaker module features four blocks in the block area. These blocks enable you to customize the sound of the Speaker module by setting its frequencies, selecting musical notes, and playing pre-programmed music.

INTERACTIONS WITH OTHER MODULES

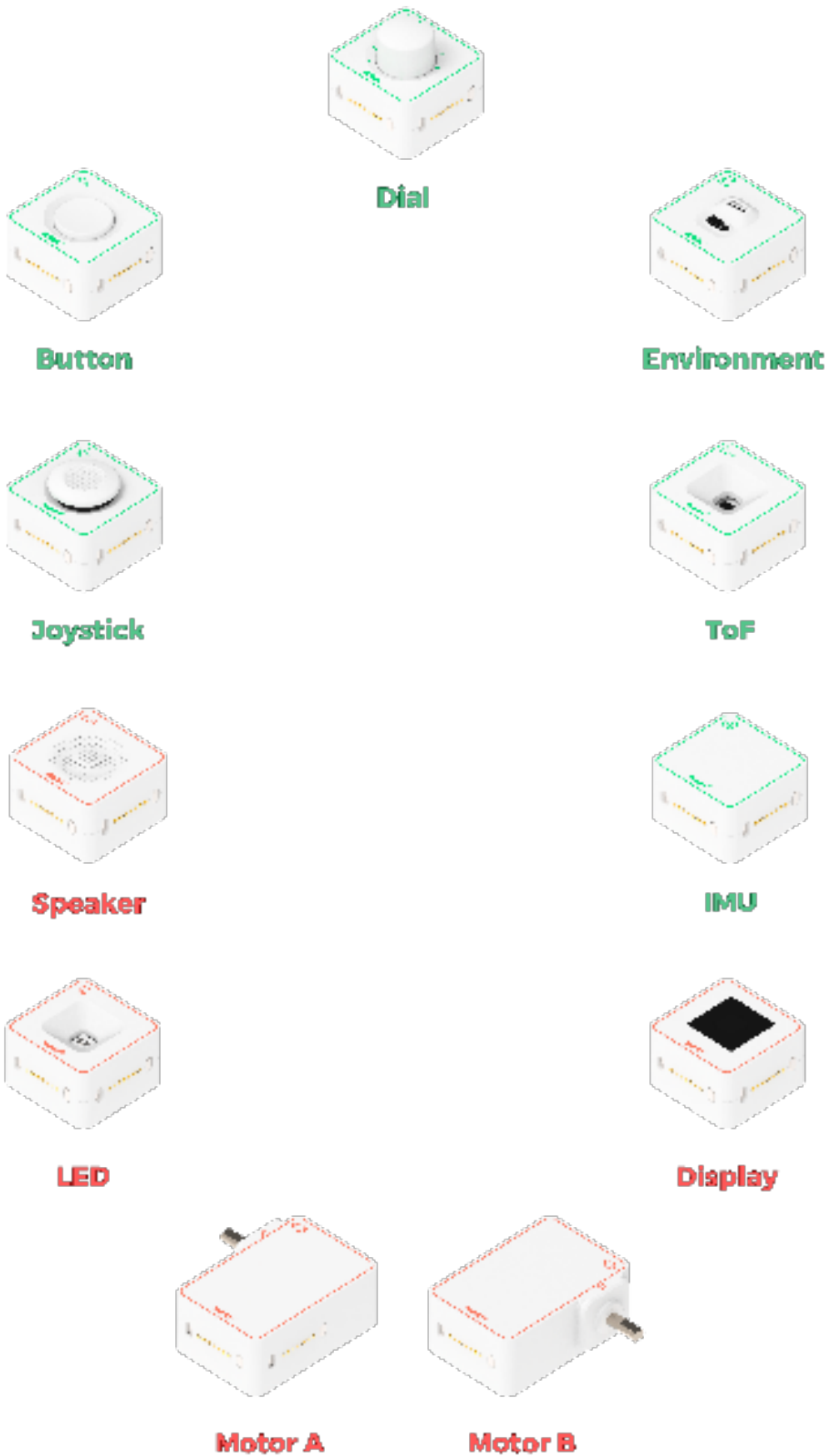
As an output module, the Speaker can interact with all input modules, producing a unique response to each. Explore these interactions by connecting the Speaker to different input modules and observe how it responds!

INTERACTION BETWEEN LED AND OTHER MODULES

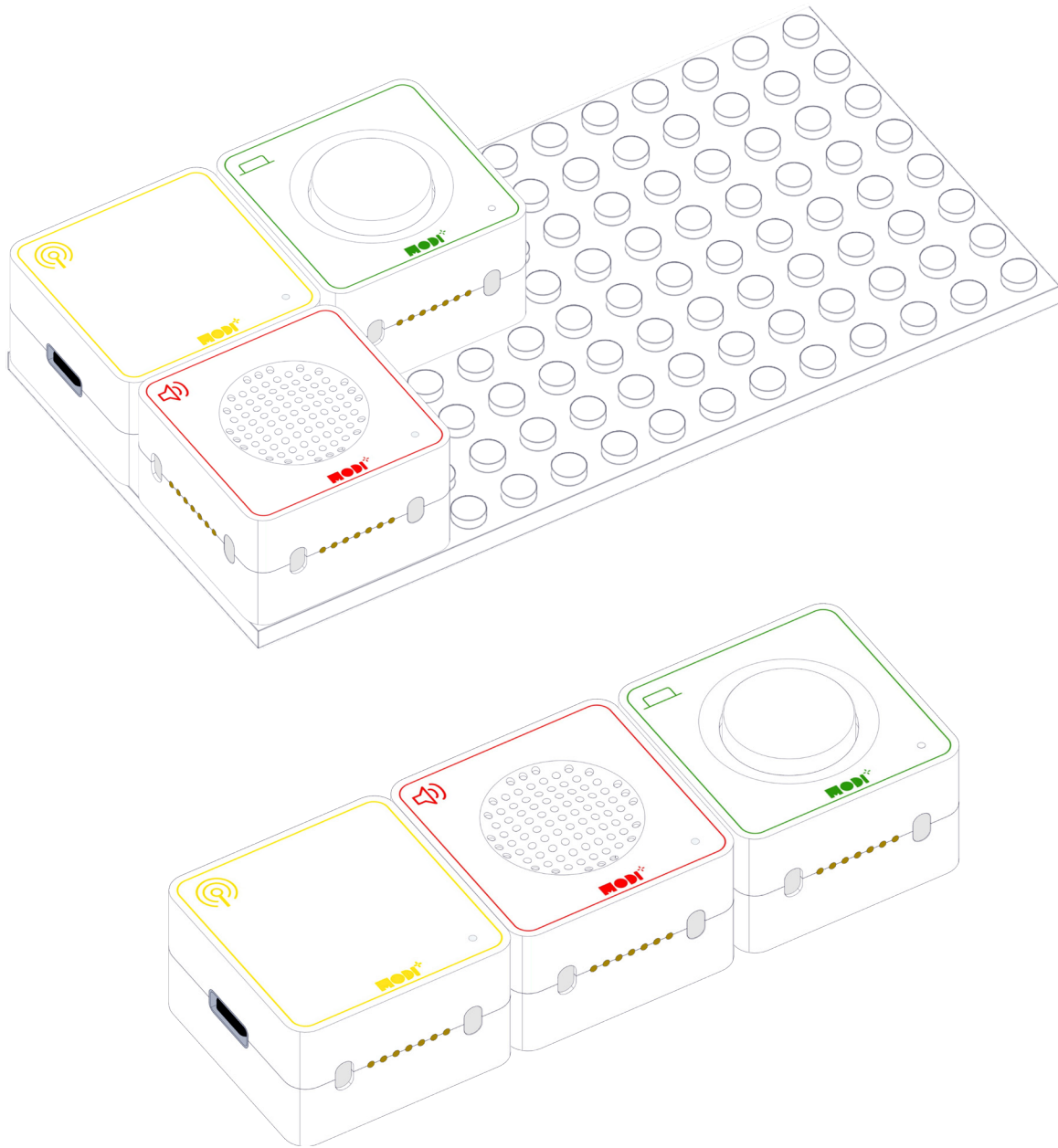


Required Modules & Brainstorming

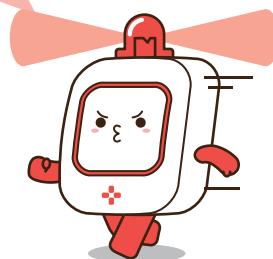
Based on the story, which MODI modules do you need for this creation?



HOW DOES THE CREATION LOOK LIKE?

**THINK ABOUT IT!**

The physical architecture you assembled does not matter. The important thing is whether you chose the right modules and whether they are working





Module Functionality

Let's look into the details of the ToF module.

1. Click



Detects if Button is clicked or not.
Range: True(Click once) : 100 / False : 0

2. Double click



Detects if Button is double - clicked or not.
Range: True(Click twice) : 100 / False : 0

3. Press



Show the press status of Button.
Range: True(Pressed) : 100 / False : 0

4. Toggle



Changes on / off when Button is clicked.
Range: True(On) : 100 / False : 0

TIPS

You can observe how input data changes in the module map on the Code Editor when you experiment with the four different click types.

